

◆ 16.11.3 Stability Integration via Lyapunov Analysis (Final — Integrated Version)

Lyapunov Functional

$$\mathcal{L}(\Psi, P) = \mathcal{E}(\Psi) + KL(P \parallel P_0)$$

where:

- $\mathcal{E}(\Psi) \geq 0$ is convex
- $KL(P \parallel P_0) \geq 0$

Thus:

$$\mathcal{L}(\Psi, P) \geq 0$$

Dissipation Inequality

Under the system dynamics defined in Section 16.11.2:

$$\begin{aligned} \dot{\mathcal{L}} &\leq \\ &- c_1 \|\nabla \mathcal{E}(\Psi)\|^2 \\ &- c_2 \|L\Psi\|^2 \\ &+ c_3 \|\nabla \ln P\|^2 \\ &+ c_4 \|u\|^2 \end{aligned}$$

$$+ C\sigma^2$$

Using boundedness assumptions:

- $\|\nabla \ln P\| \leq C$
- $\|u\| \leq u_{\max}$
- $\|L\Psi\| \geq \lambda_2 \|\Psi\|$

we obtain:

$$\begin{aligned} & \|\dot{\mathcal{L}}\| \\ & \leq \\ & -c \|\Psi\|^2 + C\sigma^2 \end{aligned}$$

Stability Implications

- Dissipativity

$\|\dot{\mathcal{L}}\| \leq 0 \quad (\text{for sufficiently small noise})$

- Mean-Square Stability

$$\begin{aligned} & \mathbb{E} \|\Psi_k\|^2 \\ & \leq \\ & (1 - c\Delta t)^k \|\Psi_0\|^2 \\ & + O(\sigma^2) \end{aligned}$$

- Ultimate Boundedness

$$\begin{aligned} & \limsup_{k \rightarrow \infty} \\ & \mathbb{E} \|\Psi_k\|^2 \\ & \leq \end{aligned}$$

$$\frac{C\sigma^2}{c}$$

- Deterministic Limit

$$\sigma \rightarrow 0$$

$$;\rightarrow;$$

$$\Psi_k \rightarrow \Psi^*$$

Additional Structural Contributions

- Network Dissipation

$$\langle \Psi, L\Psi \rangle$$

$$=$$

$$\sum_{i,j} a_{ij}(\Psi_i - \Psi_j)^2 \geq 0$$

- Control Compatibility

$$\langle \nabla \mathcal{E}(\Psi), B(u) \rangle \leq 0$$

- Discrete-Time Stability

$$\Delta t < \frac{1}{L_{\mathrm{tot}}}$$

Implication

Thus:

the system is globally stable, dissipative, and mean-square bounded, providing a Lyapunov-certified foundation for the global closure established in Section 16.11.

**SEKHEM-TSDD Theory, Mohamed Abd Elnaby,
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